LAB ASSIGNMENT – 1
DATE – 05.01.2026

NAME : Shreya Ghoshal
REGISTRATION NO. : AU/2023/0009045

ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

Q1. Create a database named collegeDB and switch to it.

```
mongosh mongodb://127.0.0.
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SHREYA GHOSHAL>mongosh
Current Mongosh Log ID: 696dd292b7edf8c5e21e2620
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
Using MongoDB:          8.2.2
Using Mongosh:          2.5.10
mongosh 2.6.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-01-16T21:22:49.705+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use collegeDB
switched to db collegeDB
```

Q2. Create a collection named students in the collegeDB database.

```
collegeDB> db.createCollection("students")
{ ok: 1 }
```

Q3. Insert one document into the students collection with the following fields:

- Name
- rollNo

- department
- age

```
collegeDB> db.students.insertOne({name:"Shreya Ghoshal", rollNo: "003", departmemt: "CSECSF", age: "20"})
{
  acknowledged: true,
  insertedId: ObjectId('696dd312b7edf8c5e21e2621')
```

Q4. Insert four more documents into the students collection with different values.

```
collegeDB> db.students.insertMany([{name:"Shreya", rollNo: "003", departmemt: "CSECSF", age: "19"}, {name:"Chayan Das", rollNo: "020", departmemt: "CSECSF",
collegeDB> db.students.insertMany([{name:"Shreya", rollNo: "003", departmemt: "CSECSF", age: "19"}, {name:"Chayan Das", rollNo: "020", departmemt: "collegeD
B> db.students.insertMany([{name:"Shreya", rollNo: "003", departmemt: "CSECSF", age: "19"}, {name:"Chayan Das", rollNo: "020", departmemt: "CSECSF", age: "2
1"}, {name:"Snigdha Sucharita Kar", rollNo: "075", departmemt: "CSEAIML", age: "21"}, {name:"Arindam Jana", rollNo: "002", departmemt: "CSECSF", age: "22"}]
)
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696dd3d1b7edf8c5e21e2622'),
    '1': ObjectId('696dd3d1b7edf8c5e21e2623'),
    '2': ObjectId('696dd3d1b7edf8c5e21e2624'),
    '3': ObjectId('696dd3d1b7edf8c5e21e2625')
  }
}
```

Q5. Display all documents present in the students collection.

```
collegeDB> db.students.find()
[
  {
    _id: ObjectId('696dd312b7edf8c5e21e2621'),
    name: 'Shreya Ghoshal',
    rollNo: '003',
    departmemt: 'CSECSF',
    age: '20'
  },
  {
    _id: ObjectId('696dd3d1b7edf8c5e21e2622'),
    name: 'Shreya',
    rollNo: '003',
    departmemt: 'CSECSF',
    age: '19'
  },
  {
    _id: ObjectId('696dd3d1b7edf8c5e21e2623'),
    name: 'Chayan Das',
    rollNo: '020',
    departmemt: 'CSECSF',
    age: '21'
  },
  {
    _id: ObjectId('696dd3d1b7edf8c5e21e2624'),
    name: 'Snigdha Sucharita Kar',
    rollNo: '075',
    departmemt: 'CSEAIML',
    age: '21'
  },
  {
    _id: ObjectId('696dd3d1b7edf8c5e21e2625'),
    name: 'Arindam Jana',
    rollNo: '002',
    departmemt: 'CSECSF',
    age: '22'
  }
]
```

Q6. Display only those students who belong to the CSE department.

```
collegeDB> db.students.find({ department: "CSECSF"})
```

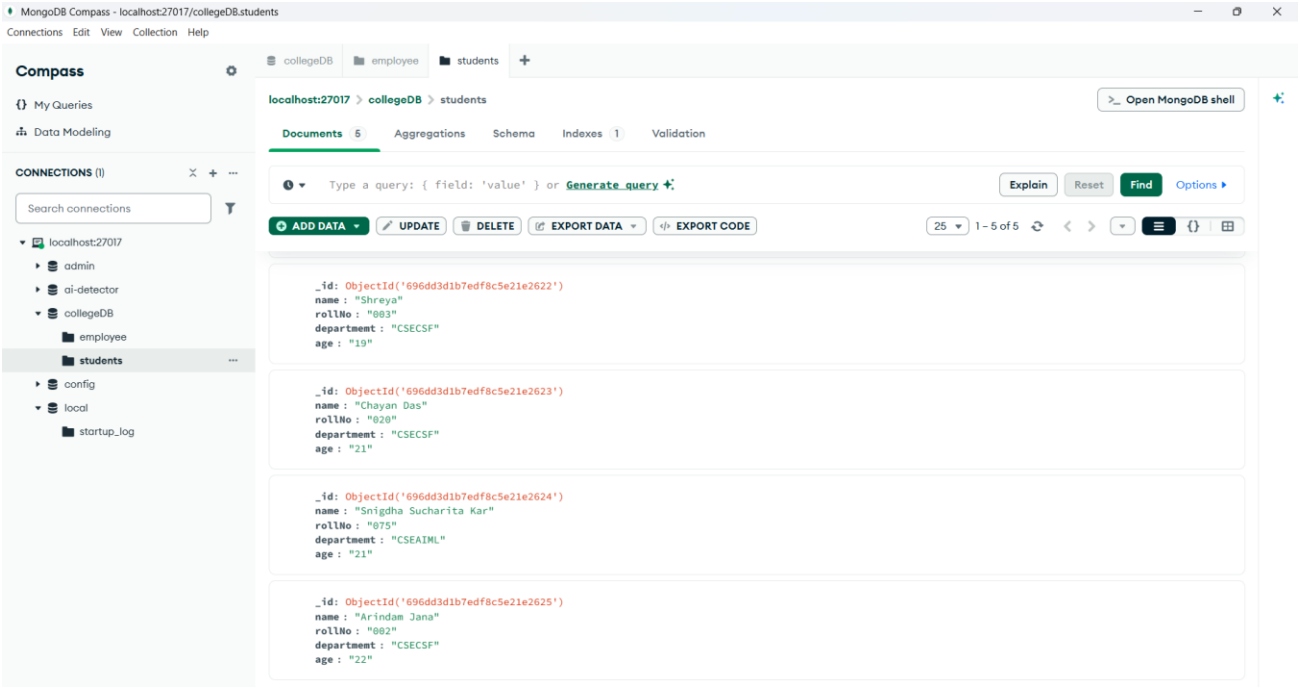
Q7. Update the age of one student (any one) to a new value.

```
db.students.updateOne({ name:"Shreya"}, {$set: {age: 21}})
ught:
axError: Unexpected token, expected "," (4:0)

|    { name: "Rahul Sharma" },
|    { $set: { age: 21 } }
| db.students.updateOne({ name:"Shreya"}, {$set: {age: 21}})
| ^
```

Q8. Delete one document from the students collection using any condition

```
collegeDB> db.students.deleteOne({ rollNo: 002 })
{ acknowledged: true, deletedCount: 0 }
collegeDB>
```


MongoDB Compass - localhost:27017/collegeDB.students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
ai-detector
collegeDB
employee
students
config
local
startup_log
collegeDB
employee
students
localhost:27017 > collegeDB > students
Open MongoDB shell
Documents 5
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
1 - 5 of 5
_id: ObjectId('696dd3d1b7edf8c5e21e2622')
name : "Shreya"
rollNo : "003"
departmemt : "CSECSF"
age : "19"
_id: ObjectId('696dd3d1b7edf8c5e21e2623')
name : "Chayan Das"
rollNo : "020"
departmemt : "CSECSF"
age : "21"
_id: ObjectId('696dd3d1b7edf8c5e21e2624')
name : "Snigdha Sucharita Kar"
rollNo : "075"
departmemt : "CSEAIML"
age : "21"
_id: ObjectId('696dd3d1b7edf8c5e21e2625')
name : "Arindam Jana"
rollNo : "002"
departmemt : "CSECSF"
age : "22"